

Research Data Quality

Every defect found in the outreach research data, what caused it, what was changed, and what the numbers look like now. All figures measured from the database, not estimated.

97% 11% FACTS SOURCED	0 155 UNSOURCED CLAIMS	8/8 0/7 EVENTS DATED	5.2 1 PAGES READ EACH	30 27 RESEARCHED	22 23 SENDABLE
--	---	---------------------------------------	--	-----------------------------------	---------------------------------

1. The one-line summary

The old pipeline fetched **one page** (the homepage), summarised it into prose, then extracted facts from that prose. Almost every defect below follows from those two decisions: a summary contains no URLs, so nothing could be sourced; and a homepage contains no placement data, so none was found.

The most dangerous single finding. B.H. Gardi College of *Engineering* was given an **A+ homoeopathy accreditation** as its email hook. The website belongs to a trust that also runs homoeopathy, nursing and ayurveda colleges, and the extractor adopted a sibling college's award as the target's. We were one send away from congratulating an engineering placement officer on an award that belongs to a different institution.

2. Every defect found

Defect	Scale	Root cause
Facts asserted with no source	155 facts only 3/27 rows had any URL	Facts were extracted from a prose summary. A summary has no URLs in it, so the model had nothing to cite and produced confidence scores from nothing. A confidence score without a source is an assertion, not evidence.
"Recent events" that were not events	0 of 7 dated 0 of 7 sourced	Homepages rarely date anything. The spec requires an event to be dated within 12 months and carry a source; none qualified, so all seven should have been <code>none_found</code> .
No placement data at all	0/27 had a rate, cohort or cell name	All of it lives on <code>/placement</code> , which was never opened.
A sibling institution's facts	Gardi (verified)	Multi-college trusts share one website. Nothing distinguished the target from its siblings.
Shared mailboxes treated as people	7 of 50 contacts	Apollo title-cases <code>placement@</code> into "Placement Ssce". Also "Training Cell", "Tpo Shegaon", "SrkR Relations", "Training Jhansi", "Tpo Institutes", "Mayank".
Boilerplate used as a hook	3 vague anchors	"Affiliated to GTU" and "approved by AICTE" are true of hundreds of colleges, so they prove nothing and read as a mail merge.

The defect found while verifying the fix

The new crawler read Gardi's placement page and extracted a **100% placement rate**. Checking the source table before trusting it:

2026 Placement Details					
Department	Registered	Placed	Placed %	Highest Package	
CIVIL	2	2	100%	2.93 LPA	<-- the "100%"
CSE/IT/CE	29	23	79%	5.00 LPA	

MBA	51	42	82.35%	7.80 LPA
MCA (2025)	70	23	32.85%	5.20 LPA

The 100% is a **two-student department**. Telling this TPO about "your 100% placement rate" would prove instantly that we had not understood their own data. These per-department, per-year tables are on almost every one of these sites, and the first number on the page is always the most flattering one.

3. What was changed

Six guards. All are enforced **in code after the model responds**, not merely requested in the prompt. An instruction is a request; code is a rule.

Guard	Where	Rule
Multi-page crawl	<code>researchCrawl.js</code>	Up to 6 pages per institution — homepage plus discovered and guessed <code>/placement</code> , accreditation, news and about pages — each with its URL retained. Junk paths (policy, privacy, approvals, grievance, fees) are excluded: they carry no facts, they push the model toward boilerplate, and they consume budget that belongs to the placement page.
Sourcing	<code>enforceSourcing()</code>	A fact whose <code>source_url</code> is not one of the pages actually fetched is demoted to 0.5 confidence — permanently below the 0.8 citation floor. Stored for audit, never citable. Invented URLs confer nothing.
Event discipline	<code>enforceSourcing()</code>	An event that is undated, unsourced, or older than ~12 months becomes <code>none_found</code> . That is a sentence about the past, not an event.
Placement rates	<code>enforceSourcing()</code>	Rejected unless <code>basis = institution_wide</code> . Also rejected when 100% sits on an unstated or tiny cohort, or the value is out of range. A per-department row is not an institution's placement rate.
Anchor specificity	<code>enforceSourcing()</code>	Affiliation and approval boilerplate with no distinguishing detail is dropped.
Entity / sibling guard	<code>enforceSourcing()</code>	<code>is_multi_institution_trust</code> and <code>sibling_institutions</code> are now extracted. A fact whose discipline conflicts with the target — homoeopathy on an engineering college — is dropped. A homoeopathy college keeps its own homoeopathy facts.
Contact identity	<code>personName.js</code>	<code>isPersonName()</code> blocks role words, single words, acronyms and blanks at the Tier 0 gate. "Dear Placement Ssce," proves the email was automated and is worse than not sending.

Tables changed

- `research_facts` — four columns added: `is_multi_institution_trust` , `sibling_institutions` , `source_urls` (every page read), `quality` (what enforcement dropped and why). Added by `ALTER TABLE ... ADD COLUMN IF NOT EXISTS` , because `CREATE TABLE IF NOT EXISTS` is a no-op on an existing table and silently adds nothing — that is what broke the first run.
- `company_campaigns` — not written by any code here. The `research_facts_sync` trigger recomputes `research_done` whenever facts change, so the flag follows the data on its own.
- Nothing else.** `company_contacts` stays read-only: the seven mailbox-contacts remain in the table and are simply blocked at send time, not deleted.

History is preserved. Re-extraction sets the previous row `is_current = false` and inserts a new version — 64 rows across 3 versions, so you can always see what was known when any email was written.

4. Quality of the research data now

Measured by the same script before and after: `npm run research:audit` .

Metric	Before	Now	
Rows with a real source URL	3/27 (11%)	29/30 (97%)	✓
Citable facts with no source	155	0	✓
Recent events carrying a date	0 of 7	8 of 8	✓

Recent events carrying a source	0 of 7	8 of 8	✓
Vague / boilerplate anchors	3	0	✓
Pages read per institution	1	5.2 avg (156 total)	✓
placement_cell_name	0%	33%	✓
top_recruiters	7%	33%	✓
publishes_placement_report	0%	20%	✓
campus_count	11%	23%	✓
naac_grade	11%	17%	✓
claimed_placement_rate	4%	3%	see below
Institutions researched	27	30	✓
Institutions sendable	23	22	see below

Two numbers went down, and both are the point.

`claimed_placement_rate` fell because per-department table rows are now rejected rather than cherry-picked. One honest institution-wide figure is worth more than thirty flattering ones.

Sendable fell by one because better research disqualifies institutions it cannot honestly personalise. B.H. Gardi now has *no hook at all* — its only candidate was boilerplate — so it is held back instead of emailed with something generic. Fewer, but defensible.

What the guards actually caught on this run

- **22 of 30** institutions were identified as multi-college trusts, so the sibling guard is live on the majority of the list, not an edge case.
- **4 events dropped** for being undated, unsourced or older than 12 months.
- **2 anchors rejected** as affiliation boilerplate.
- **0 facts demoted** for bad sourcing — because the model now cites correctly when it is given real URLs to cite. The guard is enforcement, not a crutch.

5. What this does to the emails

Regenerated from the improved facts: **17 drafts passed every validation gate, 6 were rejected, 6 were blocked before generation.** 17 distinct subject lines across 17 drafts.

An email that could not have been written before

```
Dear Akshay Mahajan,

MIT Arts, Commerce & Science College (MITACSC) recently achieved
significant recognition in the India Today Ranking 2026. In the college
segment, many student CVs face rejection by Applicant Tracking Systems
(ATS), hindering opportunities with top recruiters like Deloitte and
Amazon. At RadiusAI, our tool swiftly transforms raw student details into
ATS-parsable CVs... send us 50 CVs from your latest batch. We'll identify
which fail ATS parsing and explain the reasons, at no cost or obligation...
I would love to support the Placement Cell's efforts.
```

A **dated ranking**, two **real named recruiters** lifted from the placement page, and **their own term** for their placement team. None of that was reachable from a homepage. Every number in it — 2026, 50 — traces to the input contract.

Why the 12 that did not send, did not send

Outcome	Count	Reason
Blocked never reached the model	6	4 had no genuine hook · 3 had mailbox-names (Training Cell, Placement Ssce, Mayank) · 1 non-buyer (SevenMentor, a training company with no placement function)
Rejected written, then failed a gate	6	4 listed a fact in <code>facts_cited</code> they did not actually use · 2 rephrased the hook past the three-word overlap rule · 1 came in at 109 words against a 110 floor

Counts overlap where one email failed two gates. Every one of these is a correct rejection, not a bug: the pipeline fails closed, because a blocked email costs nothing and a bad one costs the lead.

6. Running it

Command	Does
<code>npm run research:audit</code>	Start here. Reports sourcing, hook quality, anchor specificity, contact identity and field coverage. Run it before trusting any batch.
<code>npm run facts:extract -- all --refresh</code>	Crawl and re-extract everything. Note the <code>--</code> , without it npm swallows the flag.
<code>npm run email:generate all</code>	Write and validate emails. Sends nothing.
<code>npm run email:generate -- --show <id></code>	One draft with its gates and the exact contract used.
<code>npm test</code>	58 unit tests, 21 of them new guards.

The 21 new tests each encode a defect that was actually found in this data — the Gardi homoeopathy case, the 100% two-student department, the "Placement Ssce" mailbox — so a regression fails loudly rather than quietly shipping a bad email.

7. What is still wrong

- **5 companies cannot be researched at all** — no website and no notes. There is nothing to crawl. This is a list-sourcing problem, not a code one.

- **7 contacts are shared mailboxes.** They are blocked, not fixed. Getting a named placement officer for those institutions requires new data, not better parsing.
- **18 of 30 institutions still lean on an anchor rather than a dated event.** Anchors work, but a dated event is a far stronger opening. A news source would lift reply rates more than any further prompt tuning.
- **Only 1 institution-wide placement rate survived.** Correct, but it means the pitch surface is thin. Most of these sites publish only per-department tables.
- **schema.sql is still in .gitignore**, so **research_facts**, its four new columns and every trigger exist only on this machine and in the live database. A fresh clone cannot rebuild the database.
- **The sender signature is unset.** **OUTREACH_SENDER_NAME** and its two companions are empty, so drafts end without a sign-off. These are claims about a real person, so they were left to be filled in rather than invented.